# Introduction to Robotics: Homework III

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## Introduction

The homework involves creating a kinematic simulator and controller a robotic manipulation task. Specifically, you are, first, required to identify the transformation matrices for all objects in the environment, the relative transformation matrix between the manipulator's end-effector and the object being manipulated, and the transformation matrix for the end-effector that enables the manipulated object to be precisely inserted into a designated hole. Once all the transformation matrices are determined, you are required to calculate a full end-effector trajectory in SE(3) that the robot has to follow in order to complete the manipulation. Then, you should create a kinematic simulator for the task. Finally, you need to create a task-space velocity controller (using the Jacobian pseudoinverse) that can follow the desired trajectory and complete the task in the simulator you created.

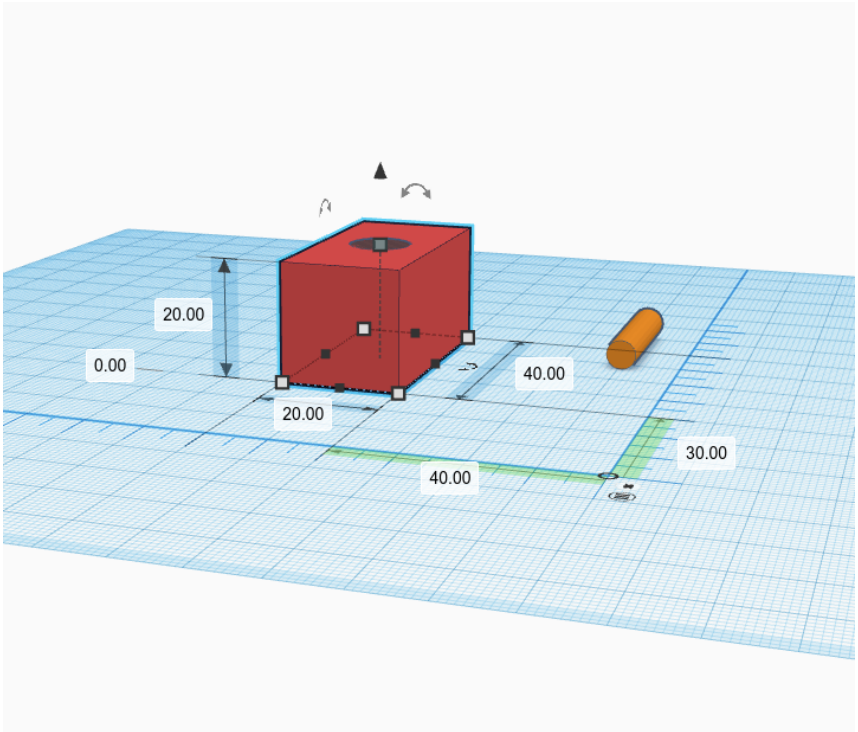


Figure 1: Visualization of the manipulation scenario. **Measurements are in cm.**

## 1 Description

The setup includes a Franka Panda manipulator equipped with a gripper (Fig. 2), a box featuring a cylindrical hole at its center (represented as the red box in Fig. 1), and a cylinder (shown as the orange cylinder in Fig. 1) that the manipulator must handle and insert into the hole. The Franka Panda is a versatile, 7-DOF robotic arm designed for precise manipulation tasks, widely used in

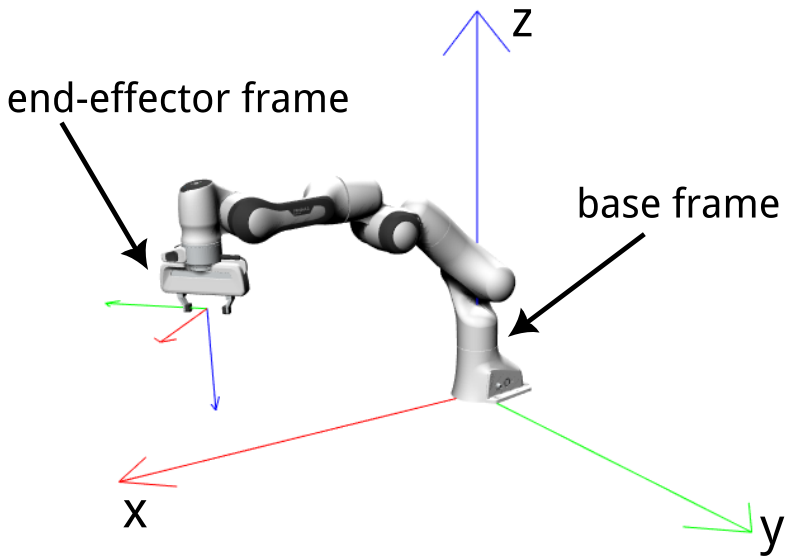


Figure 2: Visualization of the Franka Panda manipulator.

research and industrial applications due to its sensitivity and ease of control. **The kinematic structure of the robot is given in Fig. 3. The transformation from the frame $x_F$ to the end-effector is given by a rotation of $-\frac{\pi}{4}$ around the z-axis and a translation of $0.103\,m$ in the z-axis.**

## 2 Task

Given Fig. 4, Fig. 5, Fig. 6, Fig. 1, Fig. 7, Fig. 8, Fig. 2, and Fig. 3:

1. Choose the location of the world frame and assign a fixed body frame to each object;
2. Select an appropriate transformation matrix for the base frame of the manipulator with respect to the world frame, ensuring it can perform the manipulation task;
3. Determine the relative transformation matrix between the manipulator's end-effector and the cylinder, allowing the manipulator to grasp the cylinder (aka, what should be the orientation and position of the cylinder with respect to the end-effector such that it can be grasped);
4. Calculate the world frame transformation matrix of the cylinder such that, when released, it will be positioned to drop precisely into the hole in the box;
5. Design a trajectory in SE(3) such that the robot can perform the two main tasks associated with the manipulation: 1) approach the object with the correct orientation and position, and 2) once the object is grasped, move it to the correct "release" location;
6. Create a kinematic simulator that is able to kinematically represent the task;
7. Create a Jacobian pseudoinverse-based task-space velocity controller for the robot, such that it can follow the SE(3) trajectory;
8. Visualize the results using the programming language of your choice.

**Guidelines:**

- You have to use the Product of Exponentials for the solving the forward kinematics problem (if needed for your implementation);

- You have to solve the inverse kinematics problem using the Newton-Raphson algorithm (if needed for your implementation);
- You need to compute the geometric Jacobians using the exponentials algorithm;
- The designed trajectory should be continuous: you need to select appropriate waypoints, timings and use some polynomial (at least cubic) spline;
- For the simulator:
  - The simulation should operate in kinematic level: 1) there should be no dynamics involved, 2) the objects do not move (unless grasped by the robot), 3) the robot's state contains only the joint positions, 4) the robot's commands are joint velocities, 5) the robot can only grasp the cylinder, 6) once grasped (via a "grasp" command) the cylinder stays attached to the gripper until a "release" command is issued;
  - You should use an integration scheme, at minimum Euler integration;
  - The "grasp" command should only be available when close enough to the cylinder;
  - The "release" command should only be available when the cylinder is already grasped.

# 3 Deliverables

- Detailed report;
- Code for the simulator;
- Code for the trajectory generation;
- Code for the controller;
- Code for visualization.

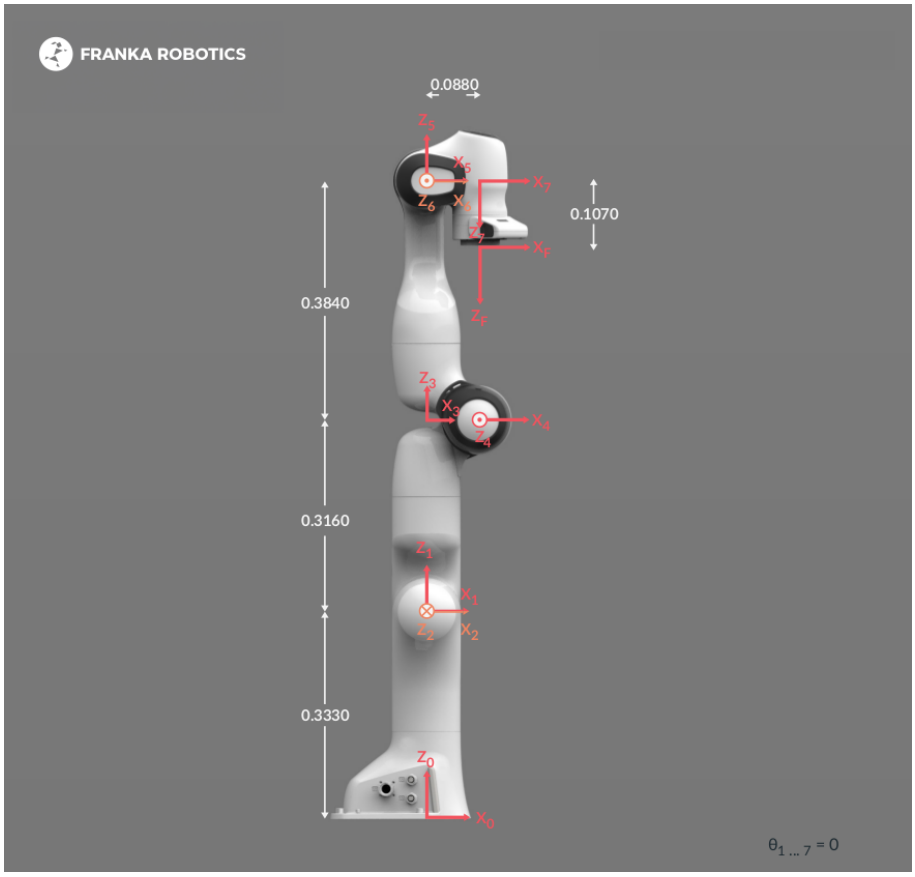


Figure 3: Franka Panda manipulator kinematic structure. **Measurements are in meters.**

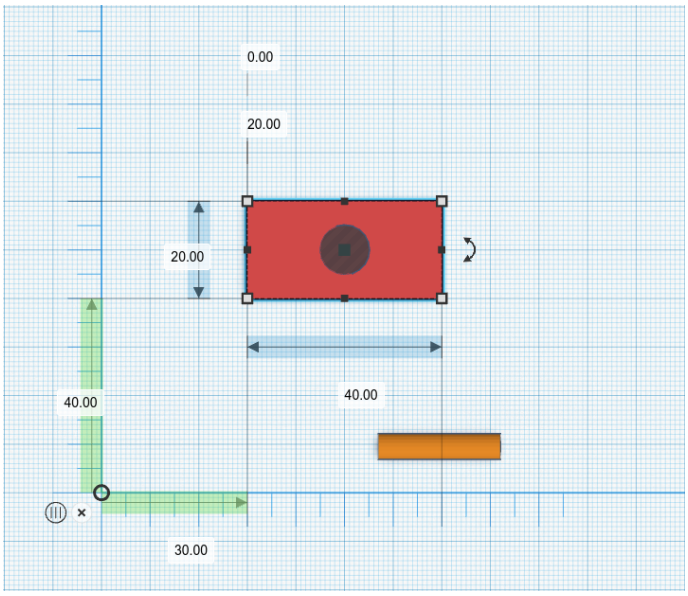


Figure 4: Scenario measurements. **Measurements are in cm.**

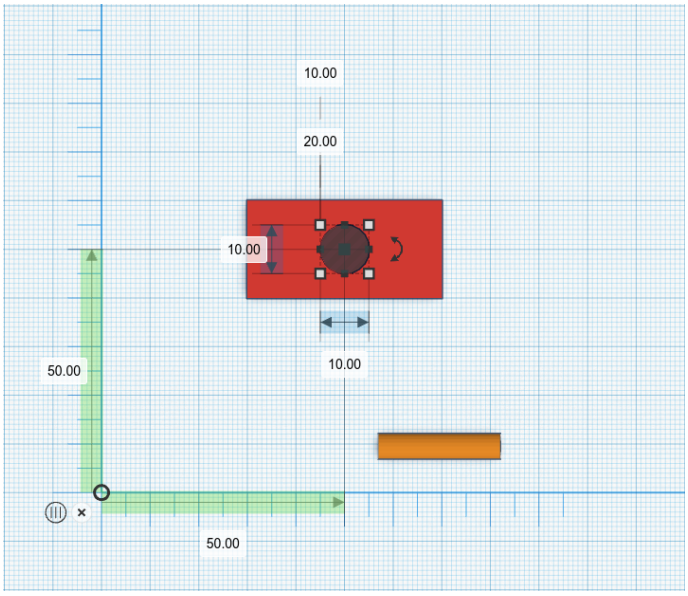


Figure 5: Scenario measurements. **Measurements are in cm.**

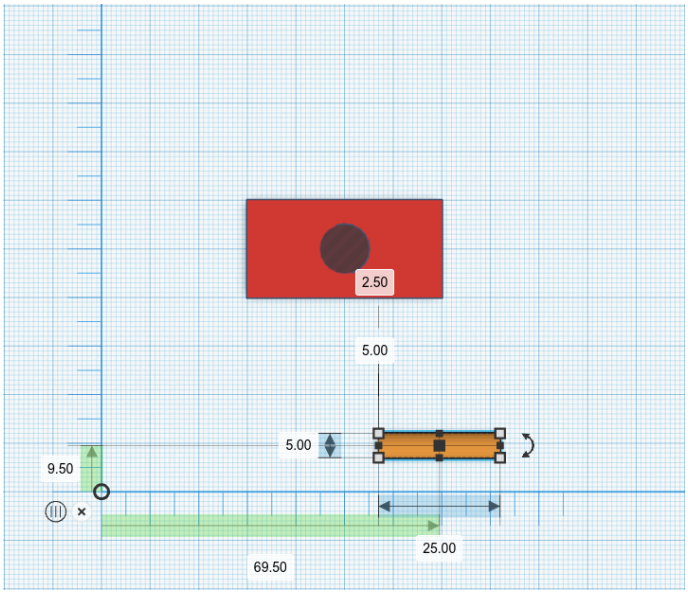


Figure 6: Scenario measurements. **Measurements are in cm.**

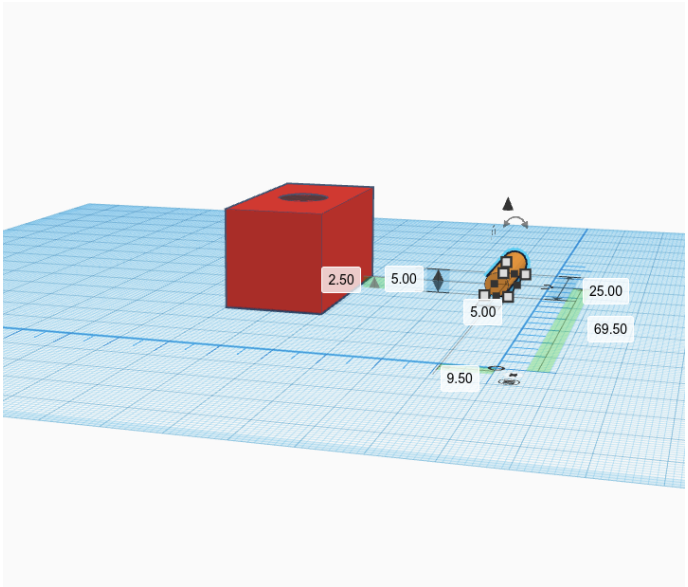


Figure 7: Scenario measurements. **Measurements are in cm.**

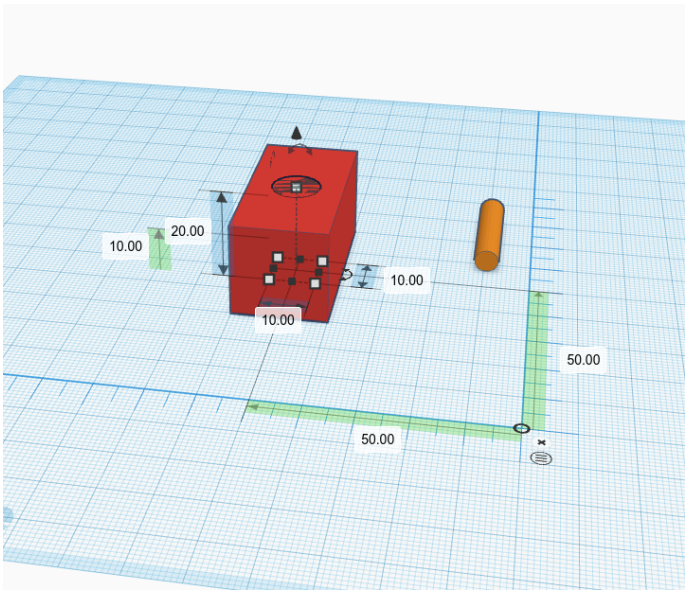


Figure 8: Scenario measurements. **Measurements are in cm.**